

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – FUNCTION CALLING & STRUCTURED OUTPUT TASK

Course Code:	CSA6502	Course Title:	Generative AI and Large Scale Models
CO Assessed:	CO2 – Precision (BL2, BL3)	Assessment:	Assessment Tool 3 – Function Calling & Structured Output Task
Weightage:	40% of CIA	Total Marks:	1 * 100 = 100 (1 Question1)
CO2 Statement:	<i>Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems (BL2, BL3)</i>		
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Section / Batch:		Date:	

Instructions to Students

- Answer 1 question. Total: 100 Marks.
- Analyze the given scenario and identify the appropriate function(s), tool(s), parameters, and structured output required to solve the task.
- Design a valid function-calling schema with appropriate function names, parameters, data types, required fields, and descriptions based on the given requirements.
- Implement the function-calling workflow demonstrating the complete process: User Query → LLM → Function/Tool Call → Function Execution → Structured Response.
- Ensure that the generated output strictly follows the defined structured format or JSON schema and contains valid, relevant, and consistent information.
- Handle appropriate edge cases such as missing parameters, invalid inputs, ambiguous requests, incorrect data types, or unavailable information, and provide suitable responses without producing invalid function calls.
- Demonstrate the working implementation using suitable test cases, including normal inputs and at least one edge-case scenario.
- Clearly document the designed schema by explaining the purpose of each function, its parameters, data types, required fields, expected input, and expected output.
- Briefly justify the design decisions and explain how function calling and structured output improve the reliability, consistency, and usability of the LLM-based application.

Submission Requirements

Students should submit:

- **Source code / Jupyter Notebook**
- **Function-calling schema**
- **Structured output/JSON schema**
- **Execution screenshots**
- **Test cases and results**
- **Brief documentation explaining the schema and function-calling workflow**

Code of Conduct

I K. Revanth Royal(192472116) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

SECTION A — FUNCTION CALLING & STRUCTURED OUTPUT TASK

16. Product Review Classification

An e-commerce company wants to classify customer reviews.

Task: Design a structured prompt that returns sentiment, product category, major complaint, and recommendation. Use suitable few-shot examples to improve classification consistency.

Also justify the technologies and techniques selected for the proposed system.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Schema Design	30%	Correctly designs a function-calling schema and structured (JSON) output format	Mostly correct schema with minor issues	Schema has significant errors	Schema is fundamentally incorrect or missing
Output Validity	25%	LLM reliably produces valid structured output matching the schema	Mostly valid output with occasional errors	Output is frequently invalid or inconsistent	Output rarely or never matches the schema
Edge-Case Handling	20%	Handles edge cases (missing fields, ambiguous input) gracefully	Handles common cases well	Handles only the simplest cases	Fails on basic cases
Function-Call Flow Demo	15%	Clear demo of the function-call to structured-response flow	Demo covers the main flow	Demo is incomplete	No functional demo presented
Schema Documentation	10%	Well-documented schema design rationale	Adequate documentation	Minimal documentation	No documentation provided

Marks Evaluation:

Question No.	Schema Design(30%)	Output Validity (25%)	Edge-Case Handling (20%)	Function-Call Flow Demo(15%)	Schema Documentation (10%)	Total Marks (100)
Q1						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Product Review Classification

An e-commerce company wants to classify customer reviews.

Objective

Design a few-shot prompt to classify customer product reviews into structured categories.

The system analyzes each review and returns:

- Sentiment
- Product Category
- Major Complaint
- Recommendation

The solution includes a structured JSON response, prompt design, schema, multiple few-shot examples, explanation of why few-shot prompting is appropriate, advantages, limitations, function-call flow, performance evaluation, and references.

Problem Analysis

An e-commerce company receives thousands of customer reviews every day for different products such as electronics, clothing, home appliances, books, and groceries.

Manually reading and categorizing these reviews is time-consuming and inconsistent. The company requires an AI system capable of automatically understanding customer feedback and generating structured information that can be used by customer support, quality assurance, and recommendation systems.

The AI should identify:

- Overall customer sentiment
- Product category
- Main complaint (if any)
- Recommended action

This structured output enables automated customer service, product quality monitoring, and business analytics.

Target Output Fields

The system extracts four important pieces of information:

- **Sentiment**
 - Positive
 - Negative
 - Neutral
- **Product Category**
 - Electronics
 - Clothing
 - Home Appliances
 - Books
 - Grocery
 - Others
- **Major Complaint**
 - Battery Issue
 - Delivery Delay
 - Damaged Product
 - Poor Quality

- Wrong Item
 - No Complaint
 - **Recommendation**
 - Refund
 - Replacement
 - Customer Support
 - No Action Needed
-

Structured JSON Schema

```
{  
  "sentiment": "Negative",  
  "product_category": "Electronics",  
  "major_complaint": "Battery Issue",  
  "recommendation": "Replacement"  
}
```

Schema Rationale

- **sentiment** identifies the customer's overall opinion.
- **product_category** classifies the reviewed product.
- **major_complaint** summarizes the primary issue.
- **recommendation** suggests the appropriate business action.

The schema provides a standardized, machine-readable format suitable for analytics dashboards, CRM platforms, and automated customer support systems.

Few-Shot Prompt

Prompt

You are an intelligent product review classifier.

Analyze every customer review and identify:

- sentiment
- product_category
- major_complaint
- recommendation

Return only valid JSON using these fields:

- sentiment
- product_category
- major_complaint
- recommendation

Example 1

Review

"The laptop battery lasts only one hour. Everything else works fine."

Output

```
{
```

```
"sentiment": "Negative",  
"product_category": "Electronics",  
"major_complaint": "Battery Issue",  
"recommendation": "Replacement"  
}
```

Example 2

Review

"The shirt fits perfectly and the fabric quality is excellent."

Output

```
{  
  "sentiment": "Positive",  
  "product_category": "Clothing",  
  "major_complaint": "No Complaint",  
  "recommendation": "No Action Needed"  
}
```

Example 3

Review

"The microwave arrived two days late, but it works perfectly."

Output

```
{  
  "sentiment": "Neutral",  
  "product_category": "Home Appliances",
```

```
"major_complaint": "Delivery Delay",  
"recommendation": "Customer Support"  
}
```

Now classify the following review.

Review

"The headphones stopped working after three days."

Expected JSON Response

```
{  
  "sentiment": "Negative",  
  "product_category": "Electronics",  
  "major_complaint": "Poor Quality",  
  "recommendation": "Replacement"  
}
```

Output Validity

Example 1

```
{  
  "sentiment": "Positive",  
  "product_category": "Books",  
  "major_complaint": "No Complaint",  
  "recommendation": "No Action Needed"  
}
```

Example 2

```
{  
  "sentiment": "Negative",  
  "product_category": "Clothing",  
  "major_complaint": "Wrong Item",  
  "recommendation": "Replacement"  
}
```

Because the output follows a fixed JSON structure, it can be directly integrated into recommendation systems, CRM software, and customer support workflows.

Why Few-Shot Prompting is Suitable

Few-shot prompting provides the language model with several representative examples before classifying new product reviews. These examples demonstrate how different review patterns should be interpreted and how the required JSON format should be generated.

It is suitable for this scenario because:

- Multiple examples teach the model to recognize different sentiments and product categories.
- It improves consistency across various review styles and writing patterns.
- The examples demonstrate how complaints should be mapped to business recommendations.
- It performs better than zero-shot and one-shot prompting for diverse customer feedback.

- It reduces ambiguity when reviews contain mixed opinions.

Thus, few-shot prompting provides higher classification accuracy and more reliable structured outputs for product review analysis.

Edge-Case Handling

Case 1: Mixed Review

Review

"The phone camera is excellent, but the battery drains quickly."

Output

```
{  
  "sentiment": "Neutral",  
  "product_category": "Electronics",  
  "major_complaint": "Battery Issue",  
  "recommendation": "Replacement"  
}
```

Case 2: Damaged Product

Review

"The washing machine arrived damaged and cannot be used."

Output

```
{  
  "sentiment": "Negative",  
  "product_category": "Home Appliances",  
  "major_complaint": "Damaged Product",  
  "recommendation": "Replacement"  
}
```

Case 3: Positive Feedback

Review

"Excellent product. Fast delivery and great quality."

Output

```
{  
  "sentiment": "Positive",  
  "product_category": "Others",  
  "major_complaint": "No Complaint",  
  "recommendation": "No Action Needed"  
}
```

Function-Call Flow

Customer Review

"The refrigerator cooling stopped working within one week."

Step 1: LLM Classification

The model analyzes the review and identifies the sentiment, product category, complaint, and recommended action.

Step 2: Structured Response

```
{  
  "sentiment": "Negative",  
  "product_category": "Home Appliances",  
  "major_complaint": "Poor Quality",  
  "recommendation": "Replacement"  
}
```

Step 3: System Action

The review is automatically tagged as **Negative**, forwarded to the customer support team, and a replacement request is initiated.

Performance Evaluation

Aspect	Few-Shot Prompting
Prompt Length	Moderate to High
Setup Effort	Moderate
Classification Accuracy	Very High
Output Consistency	Very High

Token Cost	Higher than One-Shot
Response Speed	Fast
Ease of Implementation	Easy

Advantages

- Produces highly consistent structured JSON output.
 - Handles multiple product categories effectively.
 - Improves sentiment classification accuracy.
 - Learns from several representative examples.
 - Easily integrates with CRM and recommendation systems.
 - Reduces manual review analysis.
 - Supports automated quality monitoring.
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Limitations

- Requires longer prompts than one-shot prompting.
 - Uses more tokens, increasing computational cost.
 - Performance depends on the quality and diversity of examples.
 - New product categories may require prompt updates.
 - Extremely ambiguous reviews may still require human verification.
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Conclusion

Few-shot prompting is an effective technique for product review classification because it enables the language model to learn from multiple representative examples before analyzing new reviews. These examples help the model consistently identify customer sentiment, product category, major complaints, and suitable recommendations while maintaining a structured JSON output. Compared with zero-shot and one-shot prompting, few-shot prompting offers greater reliability when handling diverse review styles and complex customer feedback.

The proposed approach allows e-commerce organizations to automate review analysis, improve customer support, monitor product quality, and generate actionable business insights. The structured JSON responses can be seamlessly integrated into recommendation engines, CRM platforms, and analytics dashboards, making few-shot prompting a practical and scalable solution for real-world product review classification.

References

1. Brown, T. et al. (2020). *Language Models are Few-Shot Learners*. NeurIPS.
2. Devlin, J. et al. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. NAACL.
3. Ouyang, L. et al. (2022). *Training Language Models to Follow Instructions with Human Feedback*. NeurIPS.
4. OpenAI. (2024). *Prompt Engineering Best Practices*.